M38 computation report

Closed meanders | Tree and bitmap geometry, four sequential families | 5 November 2026

Final exact integer

295598081822804246724487760690994952

Result	PASS; exact sum of all four family contributions
Controller wall time	3 h 26 min 27.38 s (12,387.379 s)
Peak sampled process RSS	1,489.51 GiB (1,599,349,862,400 bytes)
Work completed	4 families, 272 stages
Matrix disk I/O	0 bytes read; 0 bytes written

Family results

Family	Lower sizes	Exact contribution	Time	Peak RSS
1	1, 5, 9, 13, 17	530221880564408645809468488955515 42	0 h 47 min 05.56 s	1,168.42 GiB
2	2, 6, 10, 14, 18	773718476008210905359425367711240 28	0 h 45 min 08.58 s	1,269.71 GiB
3	3, 7, 11, 15, 19	128106091870808593177303427240780026	0 h 52 min 27.50 s	1,445.28 GiB
4	4, 8, 12, 16	370979542947336984302949477835393 56	1 h 01 min 40.19 s	1,489.51 GiB

Validation and scope

The Linux self-tests passed for 24 smaller complete orders, including the rank-64 boundary test. The M32 full-run gate reproduced 154967080111946539652848747626. All four M38 family runs passed, and their exact integer contributions sum to the reported M38. No matrix arrays were read from or written to disk; ordinary logs and source files are outside this matrix-I/O figure.

An independent CRT reconstruction of M38 has not been performed. The completed calculation and internal checks therefore do not constitute an independent mathematical proof of the algorithm's assumptions.

Provenance

Server source digest: efda251fc59d0866158215ee2e82658ed9a0a127050514f37265f0ba3eec43e4
Package SHA-256: 3dadd0f5cd99fe01ffeaba2e9a35522ac196c85d6a100dc15c158edcb6d5dea5
Runtime: Python 3.13.5, NumPy 2.3.5, Numba 0.68.0. Recorded evidence: LIVE_STATUS.json, DEPLOYMENT.json, M38_FINAL_VALIDATION.json on the server; this report was prepared from the live status receipt.